

RATCHANA POUROUCHOTTAMAN

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Master of AI graduate from Monash University with a background in computer vision and applied machine learning. During my degree I led the AI development for a sustainability project focused on rooftop greening analysis combining FastSAM, MiDaS and Stable Diffusion into a working suitability scoring pipeline - presented at a Monash industry event. I also built and deployed a live smart parking platform using Melbourne's open datasets. Every problem I worked on had real deadlines and was evaluated by academic and industry reviewers. Looking to apply my computer vision and machine learning skills to problems that have genuine real-world impact.

PROJECTS

1. Leafy Haven Rooftop Greening (City of Melbourne Sustainability Project)

- Conceived the project idea and led all AI development within a team of 6, owning the pipeline from model selection to deployment.
- Built a multi-model computer vision pipeline integrating FastSAM for rooftop segmentation, GrabCut for mask refinement, and Stable Diffusion inpainting for green visualization, with a custom HSV-based vegetation detector to assess existing green coverage.
- Designed a green factor scoring model that translated raw model outputs into interpretable rooftop suitability scores across multiple environmental dimensions.
- Deployed as a fully functional web application and presented the AI architecture at the Monash Data Science & AI industry event to academic and industry reviewers.

2. Smart Parking Availability System (Onboarding Project – City of Melbourne)

- Designed and built the MySQL database schema to integrate 6 open government datasets including ABS census data, City of Melbourne parking bay sensors and LGA boundary data.
- Led the data insights epic, analysing car ownership and population growth trends across Melbourne to surface the root causes of CBD congestion.
- Contributed to a deployed web platform providing Melbourne commuters with real-time parking availability via an interactive map interface.

3. Rift Valley Fever Clinical Trial Scheduling (MiniZinc)

- Built a multi-objective constraint optimisation model in MiniZinc to schedule treatment allocations across patient groups, balancing total treatment cost against scheduling feasibility.
- Conducted a full sensitivity analysis across 9 datasets and 10 constraint groups using the HiGHS 1.9.0 solver, systematically removing each constraint to measure its impact on solution cost and runtime.
- Found that fixed vaccination frequency was the biggest cost driver, while cost-ratio constraints caused solver timeouts in complex scenarios showing ability to reason about computational trade-offs.

4. Mine Planning Optimization (MiniZinc)

- Built a MiniZinc optimisation model to find the most profitable underground mine tunnel path across a 2D grid, maximising ore yield while staying within budget and tunnel length constraints.
- Modelled complex spatial constraints including orthogonal adjacency, unstable cell reinforcement costs, and adjacent ore collection, handling both fixed and bounded tunnel length variants.
- Solved both problem variants (fixed and bounded length), with the bounded version requiring a more flexible model where unused tunnel positions are handled dynamically.

EDUCATION

Master of Artificial Intelligence

Feb 2024 – Dec 2025

Monash University – Melbourne, VIC

Relevant Units: Machine Learning · Deep Learning · Discrete Optimisation · Planning & Automated Reasoning · Industry Experience Studio Project

Bachelor of Technology – Electronics & Communication Engineering

Aug 2019 - May 2023

Minor in Computer Science

Puducherry Technological University

TECHNICAL SKILLS

- **Core Language:** Python · SQL
- **Additional Programming:** C · C++ · JavaScript
- **ML & AI:** PyTorch · OpenCV · Ultralytics · Diffusers · Transformers · HuggingFace
- **Optimisation:** MiniZinc · Constraint Programming · HiGHS Solver
- **Web & Backend:** FastAPI · HTML · CSS · JavaScript
- **Databases:** MySQL
- **Security Tools:** Burp Suite · Wireshark · Kali Linux · Nmap · Hydra · Gobuster
- **Other:** Git · Linux · Google Workspace

WORK EXPERIENCE

Chef & Crew – Guzman Y Gomes

Dec 2025 – Present

- Progressed from casual crew to full-time chef, recognised for reliability and consistency in a high-volume kitchen environment.
- Trusted with opening and closing duties, demonstrating accountability and operational ownership beyond standard role requirements.
- Trained new staff members on food preparation standards, station procedures and food safety compliance.

Team Member – Woolworths, Clarinda

Oct 2024 – Dec 2025

- Delivered consistent customer service in a fast-paced retail environment, handling POS transactions, stock replenishment and customer enquiries.
- Maintained store presentation and safety standards in line with company requirements.

HR Trainee – Tweezerman India Pvt. Ltd.

Jul 2023 – Dec 2023

- Supported HR operations including recruitment coordination, employee record management and administrative reporting.
- Handled sensitive employee information with confidentiality and accuracy.

INTERNSHIP

Cybersecurity and Ethical Hacking internship

Jul 2021 – Sep 2021

RINEX Technologies, in collaboration with E-Cell, IIT Kharagpur

- Completed hands-on penetration testing across two projects - a network hackathon and web application security assessment using industry-standard tools on Kali Linux.
- Performed network reconnaissance, port scanning, brute-force attacks and privilege escalation using Nmap, Hydra and Gobuster to identify and exploit vulnerabilities in a controlled environment.
- Conducted web application security testing identifying vulnerabilities including SQL injection, CSRF, clickjacking, sensitive data exposure and security misconfiguration using Burp Suite and Wireshark.

LANGUAGES

English(Fluent)

Tamil(Native)